

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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NET 323

Digging into AWS network performance

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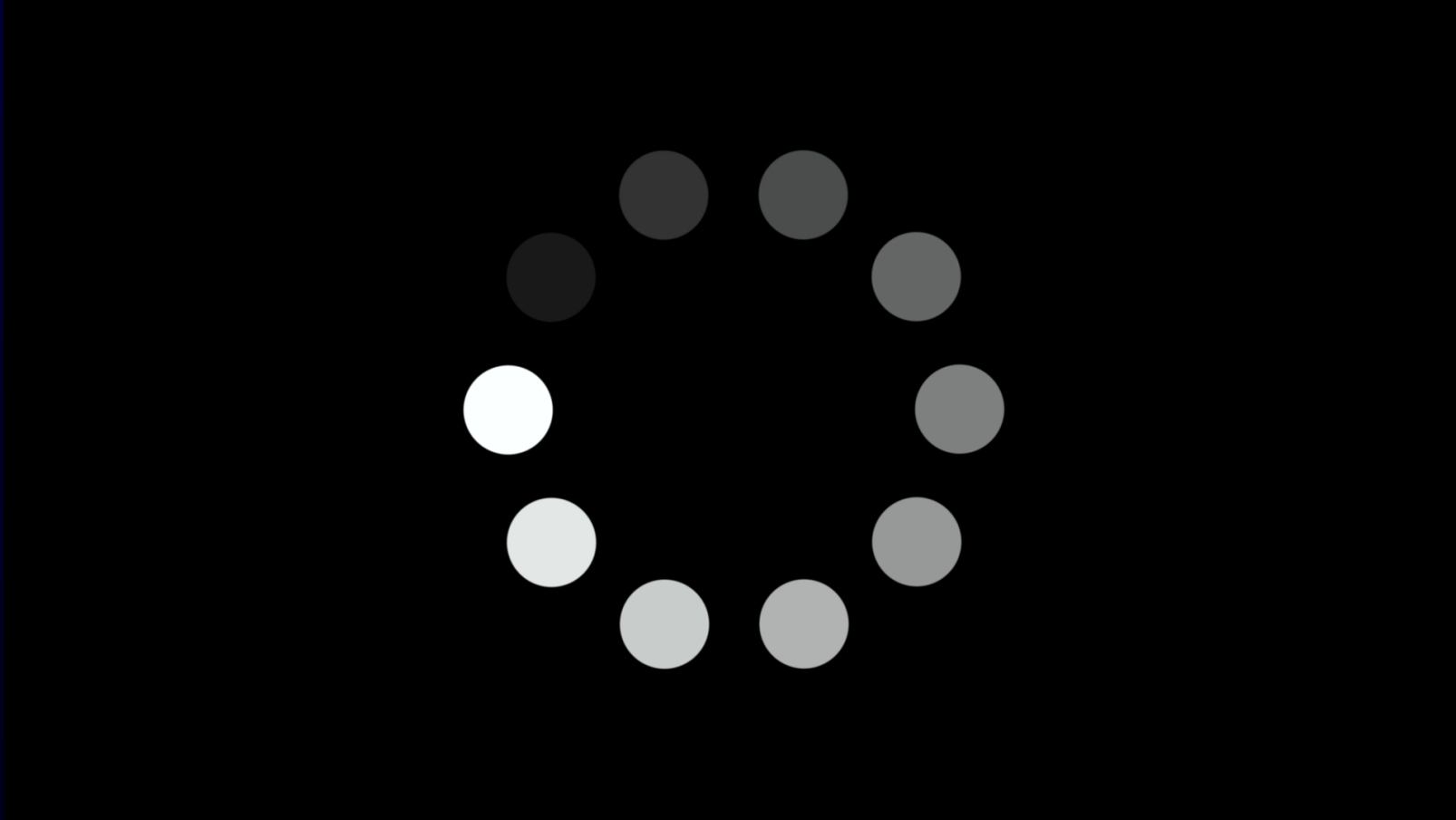
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Slow is the new broken



Network performance concepts



Bandwidth

Maximum rate at which data can be transferred



Throughput

Actual data rate at which data is transferred



Latency

Amount of time it takes for data to travel from one point to another

Amazon EC2 bandwidth

Instance network bandwidth depends on the number of vCPUs that it has

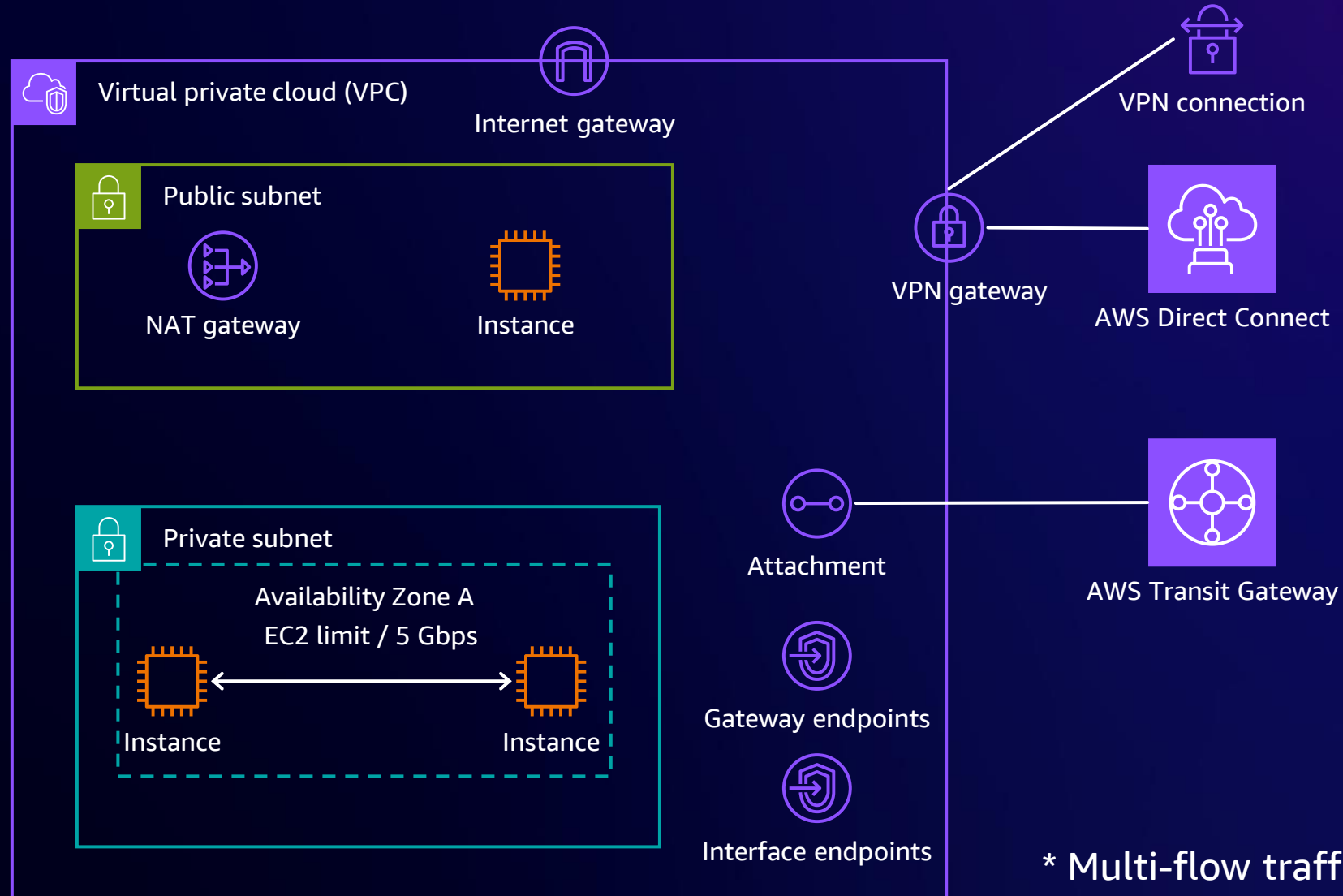
Instance bandwidth specifications apply to both inbound and outbound

However, **there is more...**

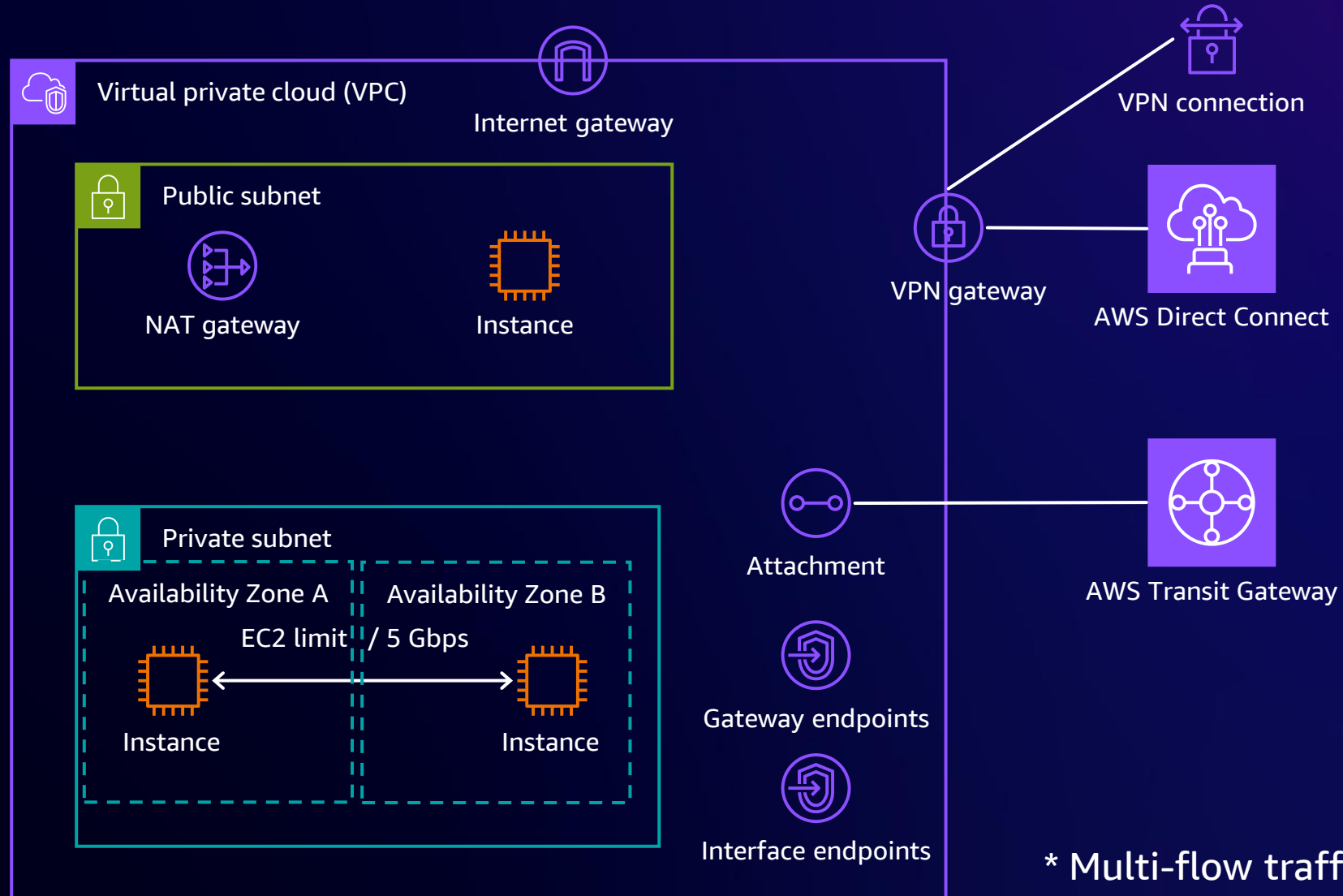
DescribeInstanceTypes	
c5.4xlarge	Up to 10 Gigabit
c5.xlarge	Up to 10 Gigabit
c5.12xlarge	12 Gigabit
c5.24xlarge	25 Gigabit
c5.9xlarge	10 Gigabit
c5.2xlarge	Up to 10 Gigabit
c5.large	Up to 10 Gigabit
c5.metal	25 Gigabit
c5.18xlarge	25 Gigabit

Instance type	Baseline bandwidth (Gbps)	Burst bandwidth (Gbps)
c5.large	0.75	10.0
c5.xlarge	1.25	10.0
c5.2xlarge	2.5	10.0
c5.4xlarge	5.0	10.0

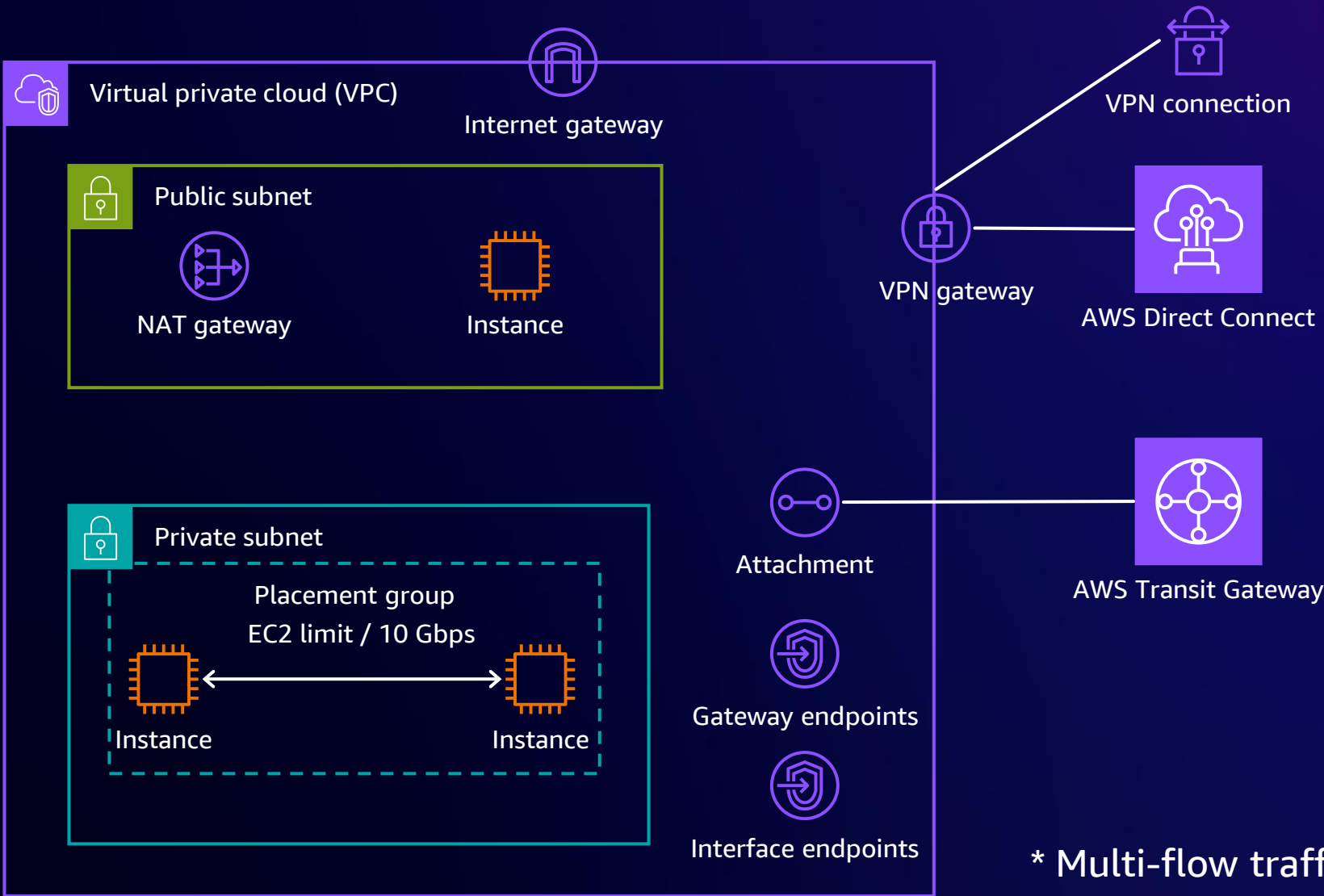
Bandwidth limits



Bandwidth limits

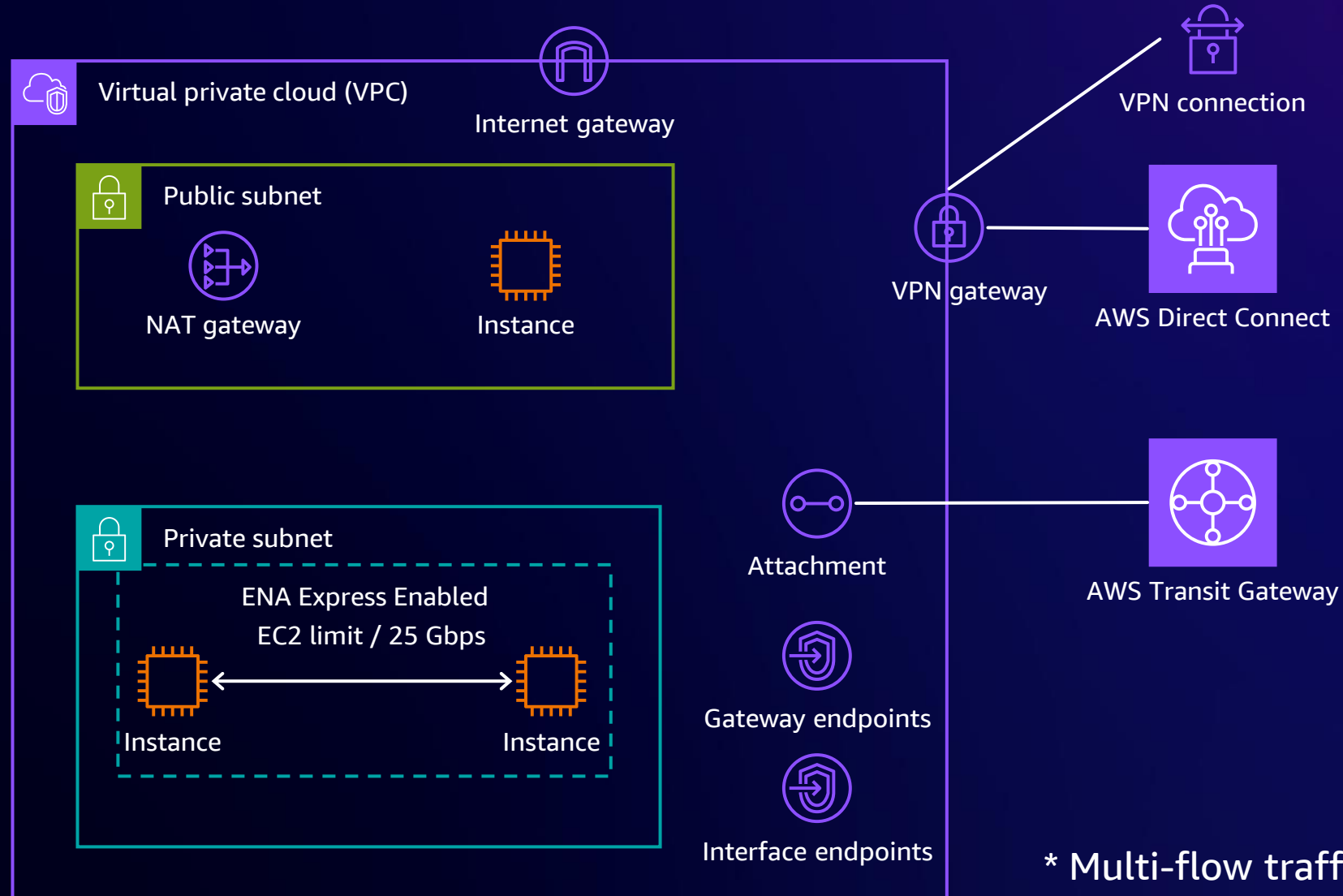


Bandwidth limits

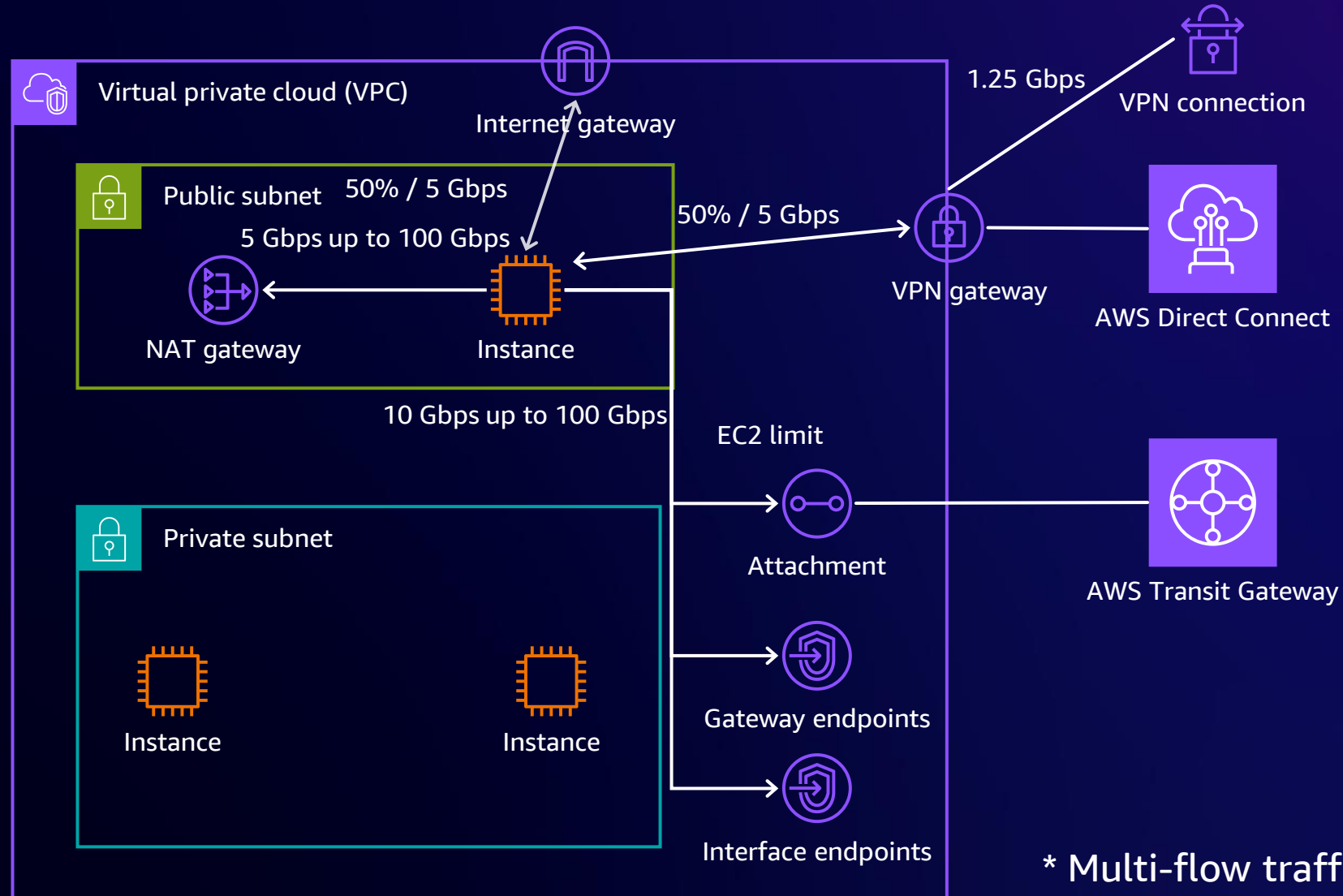


* Multi-flow traffic/single-flow traffic

Bandwidth limits



Bandwidth limits



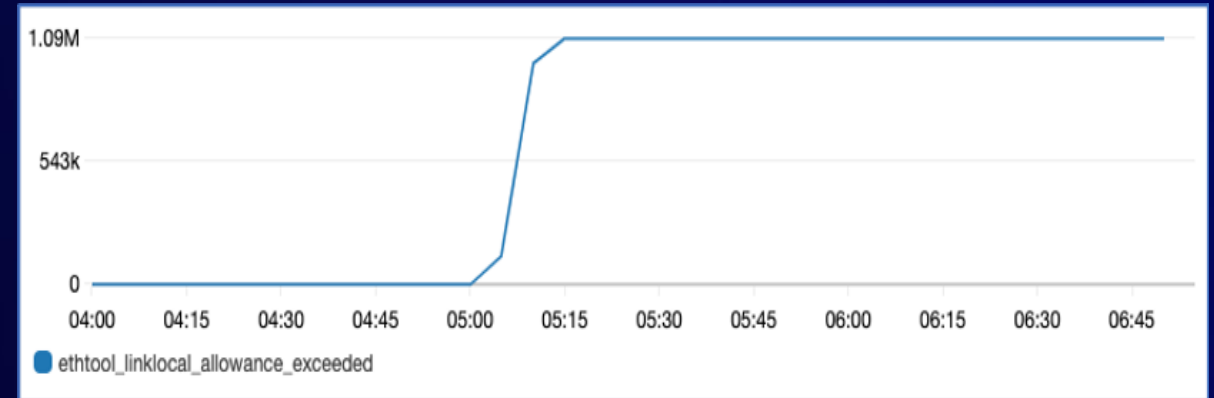
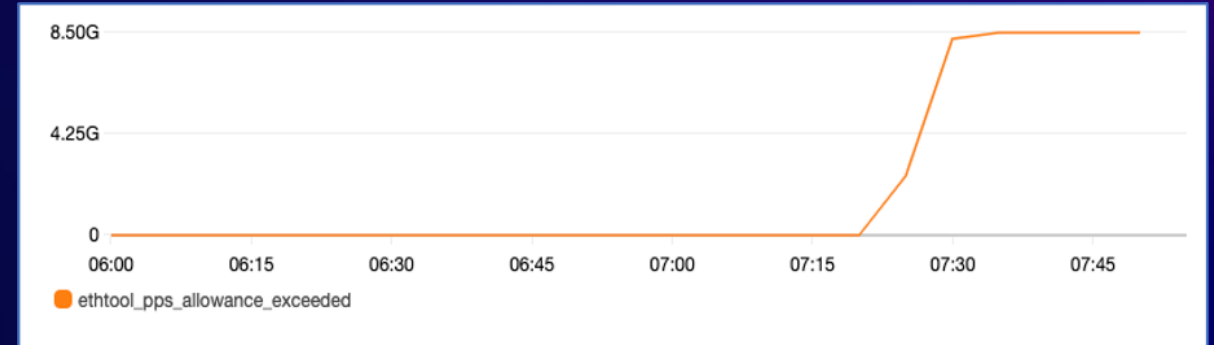
* Multi-flow traffic/single-flow traffic

Monitoring network performance

The Elastic Network Adapter (ENA) driver publishes metrics

- Inbound and outbound bandwidth
- Packets per second (PPS)
- PPS to link-local services
- Connections tracked (conntrack)

Use these to **monitor, troubleshoot, and rightsize** your workloads



Testing network performance

Actual performance can be influenced by many factors

- Instance family, type, size
- Application profile (packet size, protocol)
- Instance utilization (CPU, memory, disk)
- Latency (users and workloads)

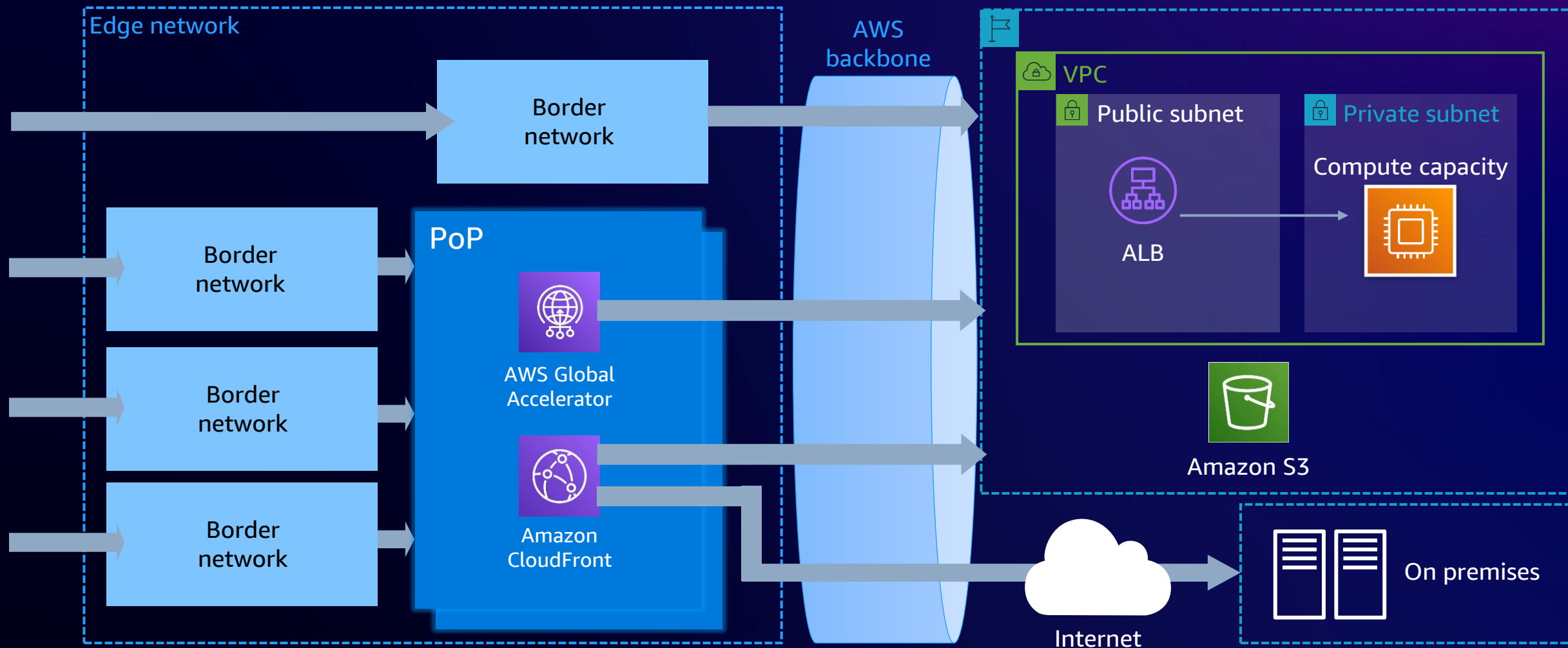
We recommend you **test it yourself**



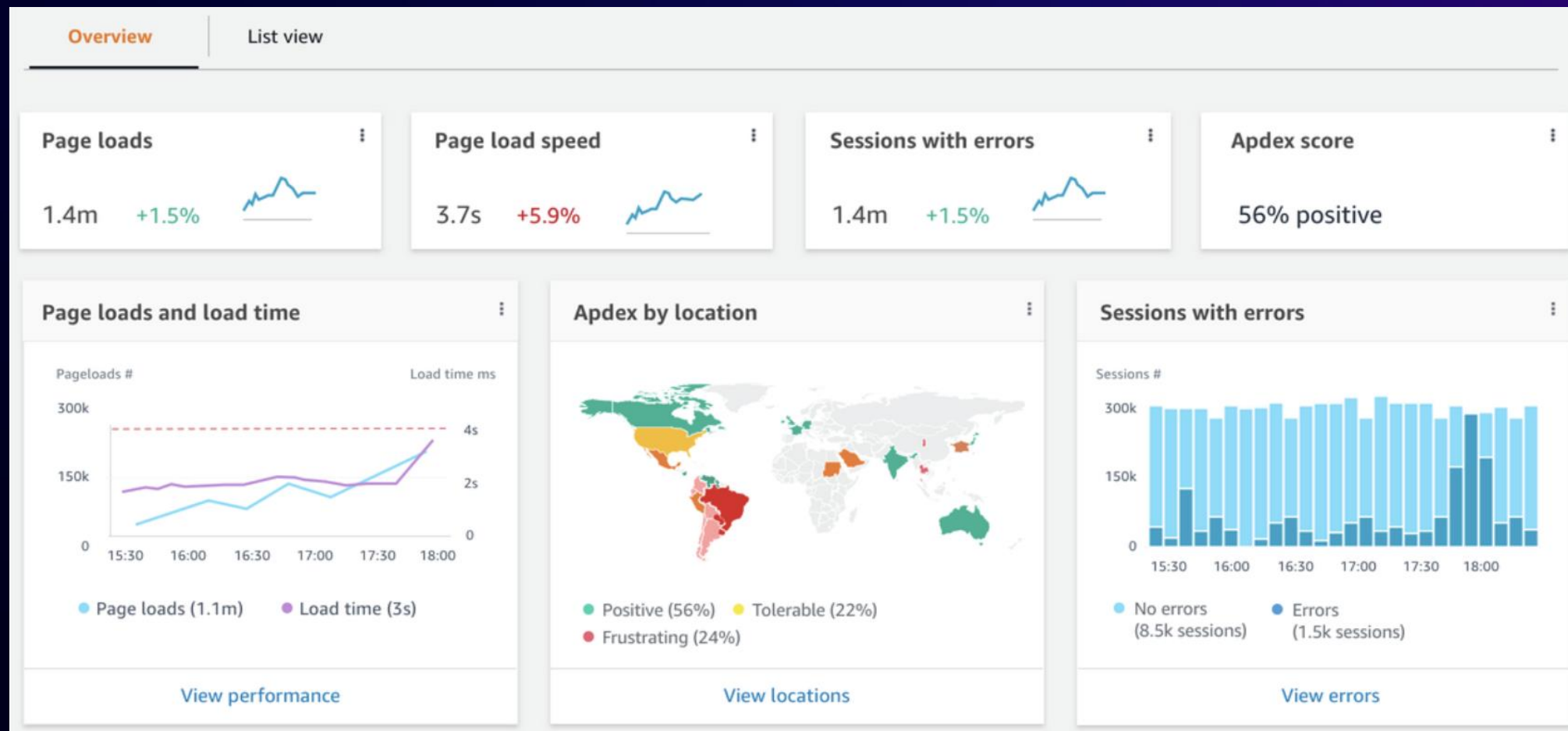
[Benchmark network throughput between EC2 instances](#)

Guide on how to measure network performance inside an Amazon VPC

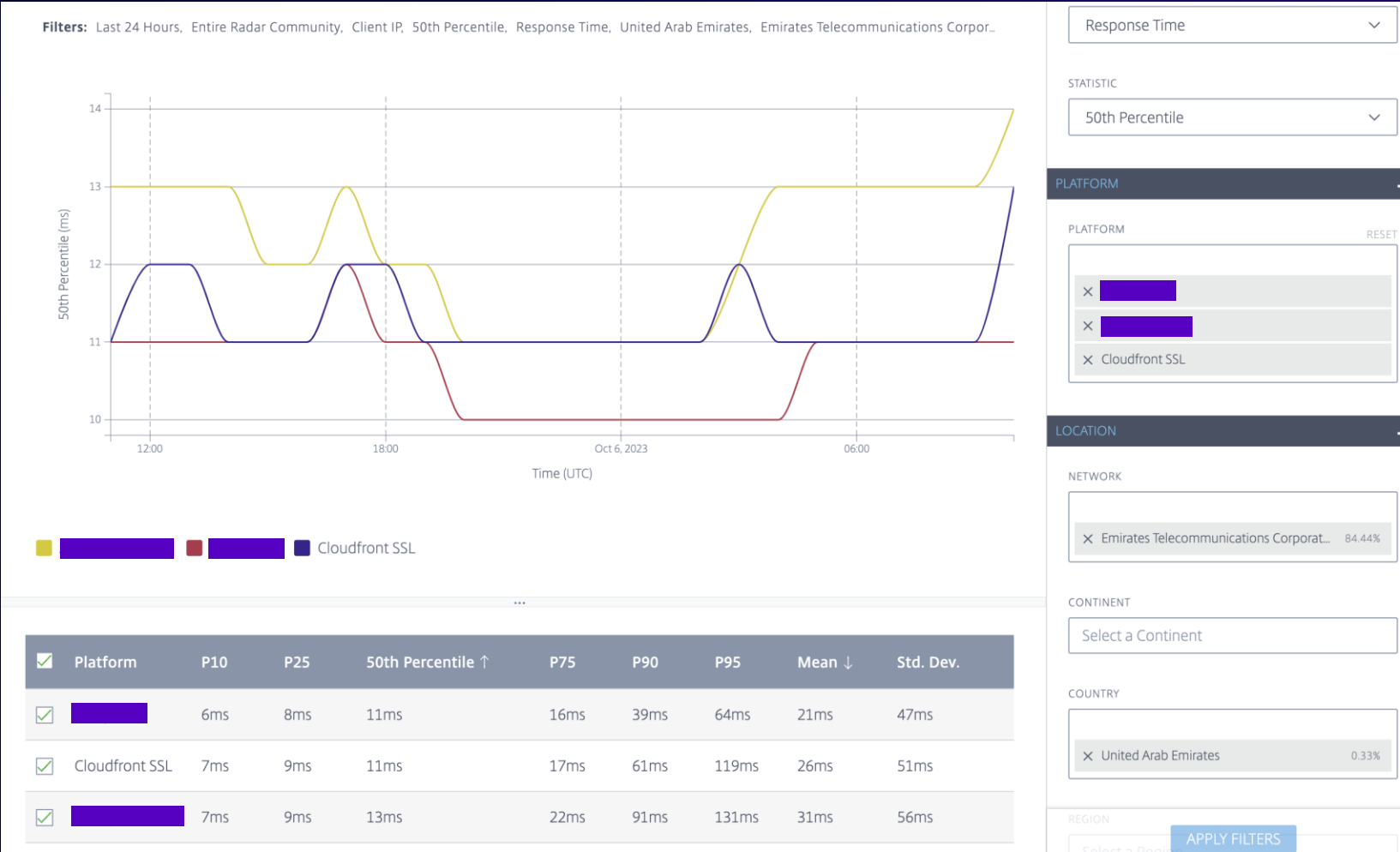
AWS edge services



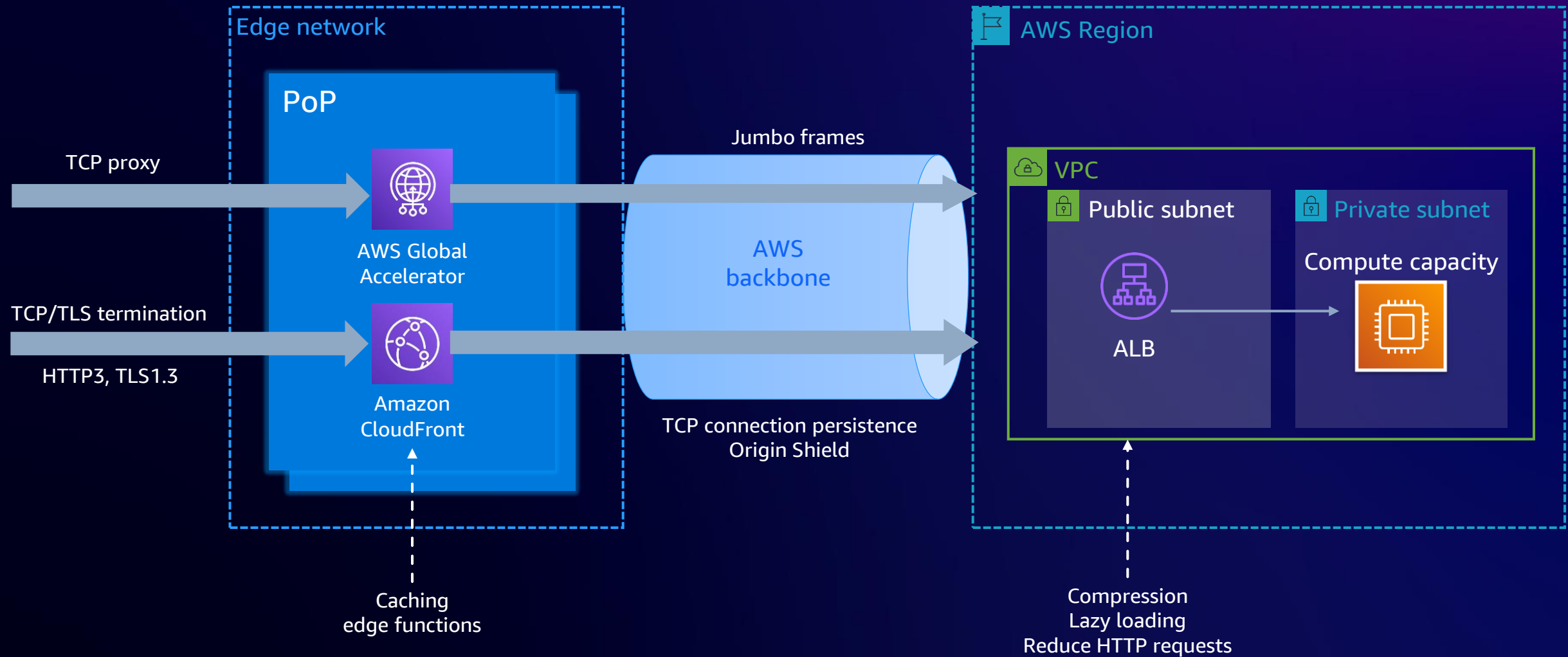
Measuring performance at the edge – RUM



Measuring performance at the edge – RUM



Improve performance at the edge



Let's start the conversation!

- Tell us about a performance tweak that you are proud of
- What is your main challenge in measuring performance?
- What tools are you currently using for testing?

Thank you!



Please complete the session survey in the mobile app

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